

Algebra II with Trigonometry

Chapter 6.4

Olympiad Solutions (GPT-5.4)

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[4533650_alg-2trig-h-chp-6-4-note.pdf](#)

Solutions

1. Find all real numbers x in $[0, 2\pi)$ such that

$$\tan x + \cot x = 2 \sec 2x.$$

We first note the equation is defined only when $\sin x \neq 0$, $\cos x \neq 0$, and $\cos 2x \neq 0$.

Now

$$\tan x + \cot x = \frac{\sin x}{\cos x} + \frac{\cos x}{\sin x} = \frac{\sin^2 x + \cos^2 x}{\sin x \cos x} = \frac{1}{\sin x \cos x}.$$

Also,

$$\sin 2x = 2 \sin x \cos x,$$

so

$$\tan x + \cot x = \frac{2}{\sin 2x}.$$

Thus the equation becomes

$$\frac{2}{\sin 2x} = 2 \sec 2x = \frac{2}{\cos 2x}.$$

Since both sides are defined, we may divide by 2:

$$\frac{1}{\sin 2x} = \frac{1}{\cos 2x},$$

hence

$$\sin 2x = \cos 2x.$$

Therefore

$$\tan 2x = 1,$$

so

$$2x = \frac{\pi}{4} + n\pi.$$

Hence

$$x = \frac{\pi}{8} + \frac{n\pi}{2}.$$

Now restrict to $[0, 2\pi)$. Taking $n = 0, 1, 2, 3$, we get

$$x = \frac{\pi}{8}, \frac{5\pi}{8}, \frac{9\pi}{8}, \frac{13\pi}{8}.$$

So the solutions are

$$\boxed{\frac{\pi}{8}, \frac{5\pi}{8}, \frac{9\pi}{8}, \frac{13\pi}{8}}.$$

2. Let

$$f(x) = \sec x + \csc x.$$

Find the minimum value of $f(x)$ over all real x for which it is defined.

We seek the minimum of

$$f(x) = \frac{1}{\cos x} + \frac{1}{\sin x},$$

where $\sin x \neq 0$ and $\cos x \neq 0$.

If $\sin x$ and $\cos x$ have opposite signs, then one term is positive and the other negative; near points where one denominator approaches 0, $f(x)$ can become arbitrarily negative. For example, as $x \rightarrow \pi^+$,

$$\sin x \rightarrow 0^-, \quad \cos x \rightarrow -1,$$

so

$$\csc x \rightarrow -\infty, \quad \sec x \rightarrow -1,$$

hence

$$f(x) \rightarrow -\infty.$$

Therefore $f(x)$ is not bounded below on its domain, so it has no minimum value.

Thus the answer is:

No minimum exists; $f(x)$ is unbounded below.

3. How many vertical asymptotes does the graph of

$$y = \tan(2x) + \cot(3x)$$

have on the interval $(0, \pi)$?

Vertical asymptotes occur where either $\tan(2x)$ or $\cot(3x)$ is undefined.

From $\tan(2x)$:

$$\cos(2x) = 0 \implies 2x = \frac{\pi}{2} + n\pi$$

so

$$x = \frac{\pi}{4} + \frac{n\pi}{2}.$$

In $(0, \pi)$, this gives

$$x = \frac{\pi}{4}, \frac{3\pi}{4}.$$

From $\cot(3x)$:

$$\sin(3x) = 0 \implies 3x = n\pi$$

so

$$x = \frac{n\pi}{3}.$$

In $(0, \pi)$, this gives

$$x = \frac{\pi}{3}, \frac{2\pi}{3}.$$

These four points are distinct, so there are

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vertical asymptotes.

4. Determine all real x satisfying

$$\sec x - \tan x = \csc x - \cot x.$$

Assume all expressions are defined, so $\sin x \neq 0$ and $\cos x \neq 0$.

Use the standard identities:

$$\sec x - \tan x = \frac{1 - \sin x}{\cos x}, \quad \csc x - \cot x = \frac{1 - \cos x}{\sin x}.$$

So the equation becomes

$$\frac{1 - \sin x}{\cos x} = \frac{1 - \cos x}{\sin x}.$$

Cross-multiply:

$$\sin x(1 - \sin x) = \cos x(1 - \cos x).$$

Thus

$$\sin x - \sin^2 x = \cos x - \cos^2 x.$$

Rearrange:

$$\sin x - \cos x = \sin^2 x - \cos^2 x.$$

Factor the right-hand side:

$$\sin x - \cos x = (\sin x - \cos x)(\sin x + \cos x).$$

Hence

$$(\sin x - \cos x)(1 - (\sin x + \cos x)) = 0.$$

So either

Case 1:

$$\sin x = \cos x.$$

Then

$$\tan x = 1 \implies x = \frac{\pi}{4} + n\pi.$$

Case 2:

$$\sin x + \cos x = 1.$$

Square:

$$(\sin x + \cos x)^2 = 1$$

gives

$$1 + 2 \sin x \cos x = 1,$$

so

$$\sin x \cos x = 0.$$

But then $\sin x = 0$ or $\cos x = 0$, where the original equation is undefined. So this case yields no valid solutions.

Therefore all solutions are

$$x = \frac{\pi}{4} + n\pi \quad (n \in \mathbb{Z}).$$

5. A function of the form

$$y = a \tan(k(x - b))$$

has consecutive vertical asymptotes at $x = \frac{\pi}{12}$ and $x = \frac{5\pi}{12}$, and its graph passes through $(\frac{\pi}{4}, -3)$. Find a , k , and one possible value of b .

For $y = a \tan(k(x - b))$, consecutive vertical asymptotes are separated by

$$\frac{\pi}{k}.$$

The given asymptotes are at

$$\frac{\pi}{12}, \quad \frac{5\pi}{12},$$

whose distance is

$$\frac{5\pi}{12} - \frac{\pi}{12} = \frac{\pi}{3}.$$

Hence

$$\frac{\pi}{k} = \frac{\pi}{3} \implies k = 3.$$

The midpoint between consecutive asymptotes is

$$\frac{\frac{\pi}{12} + \frac{5\pi}{12}}{2} = \frac{\pi}{4}.$$

At this midpoint, the tangent argument is 0, so the graph crosses the midline $y = 0$. But the given point is $(\frac{\pi}{4}, -3)$, not on the midline. So we should instead determine b from the asymptote formula directly.

Vertical asymptotes occur when

$$k(x - b) = \frac{\pi}{2} + n\pi.$$

With $k = 3$, this gives

$$3(x - b) = \frac{\pi}{2} + n\pi.$$

Take $x = \frac{\pi}{12}$:

$$3\left(\frac{\pi}{12} - b\right) = \frac{\pi}{2} + n\pi.$$

A convenient choice is $n = -1$, giving

$$3\left(\frac{\pi}{12} - b\right) = -\frac{\pi}{2}.$$

So

$$\frac{\pi}{4} - 3b = -\frac{\pi}{2} \implies 3b = \frac{3\pi}{4} \implies b = \frac{\pi}{4}.$$

This indeed gives asymptotes at

$$x = \frac{\pi}{4} + \frac{\pi/2 + n\pi}{3} = \frac{\pi}{4} + \frac{\pi}{6} + \frac{n\pi}{3} = \frac{5\pi}{12} + \frac{n\pi}{3},$$

which includes both $\frac{\pi}{12}$ and $\frac{5\pi}{12}$.

Now use the point $(\frac{\pi}{4}, -3)$:

$$y = a \tan\left(3\left(\frac{\pi}{4} - \frac{\pi}{4}\right)\right) = a \tan 0 = 0,$$

which cannot equal -3 .

So there is an inconsistency in the stated data: any function $a \tan(k(x - b))$ with those consecutive asymptotes must have its zero at the midpoint $x = \frac{\pi}{4}$, and therefore cannot pass through $(\frac{\pi}{4}, -3)$.

Hence:

No such function exists.

(From the asymptotes alone one gets $k = 3$ and one possible $b = \frac{\pi}{4}$, but then the point condition is impossible.)

6. Find the exact value of

$$\sec \frac{\pi}{12} \csc \frac{5\pi}{12} + \sec \frac{5\pi}{12} \csc \frac{\pi}{12}.$$

Write everything in terms of sine and cosine:

$$\sec \frac{\pi}{12} \csc \frac{5\pi}{12} = \frac{1}{\cos \frac{\pi}{12} \sin \frac{5\pi}{12}},$$

and

$$\sec \frac{5\pi}{12} \csc \frac{\pi}{12} = \frac{1}{\cos \frac{5\pi}{12} \sin \frac{\pi}{12}}.$$

Now use

$$\sin \frac{5\pi}{12} = \cos \frac{\pi}{12}, \quad \cos \frac{5\pi}{12} = \sin \frac{\pi}{12},$$

since $\frac{5\pi}{12} = \frac{\pi}{2} - \frac{\pi}{12}$.

Thus the expression becomes

$$\frac{1}{\cos^2 \frac{\pi}{12}} + \frac{1}{\sin^2 \frac{\pi}{12}} = \sec^2 \frac{\pi}{12} + \csc^2 \frac{\pi}{12}.$$

Combine:

$$\frac{1}{\cos^2 \theta} + \frac{1}{\sin^2 \theta} = \frac{\sin^2 \theta + \cos^2 \theta}{\sin^2 \theta \cos^2 \theta} = \frac{1}{\sin^2 \theta \cos^2 \theta},$$

where $\theta = \frac{\pi}{12}$.

Since

$$\sin 2\theta = 2 \sin \theta \cos \theta,$$

we get

$$\sin^2 \theta \cos^2 \theta = \frac{\sin^2 2\theta}{4}.$$

Hence

$$\frac{1}{\sin^2 \theta \cos^2 \theta} = \frac{4}{\sin^2 2\theta}.$$

Now $2\theta = \frac{\pi}{6}$, so

$$\sin \frac{\pi}{6} = \frac{1}{2}.$$

Therefore

$$\frac{4}{\sin^2 \frac{\pi}{6}} = \frac{4}{(1/2)^2} = 16.$$

So the exact value is

$$\boxed{16}.$$